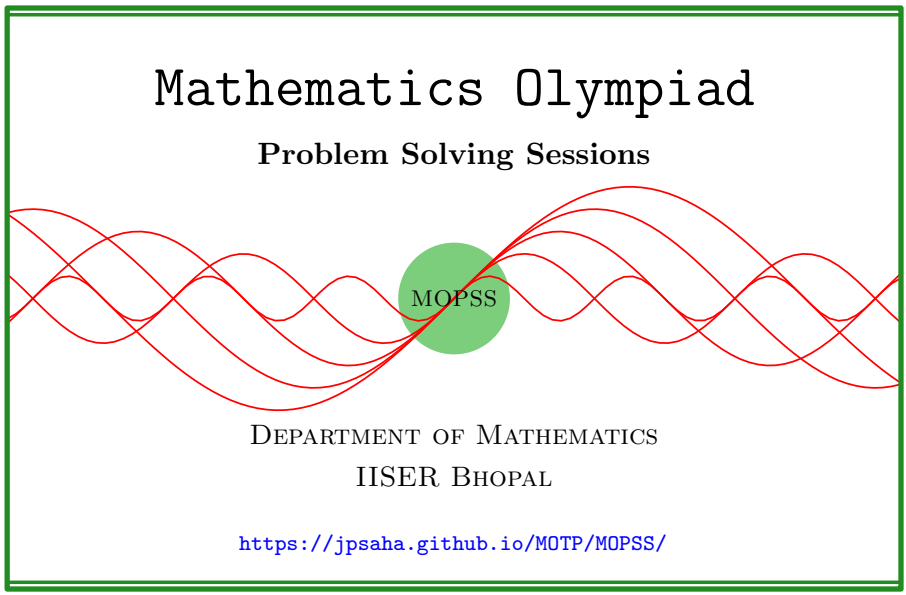
Triangles

MOPSS



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

List of problems and examples

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§1 Triangles

Exercise 1 (RMO 1991 P5,). Take any point P_1 on the side BC of a triangle ABC and draw the following chain of lines: P_1P_2 parallel to AC (P_2 on AB), P_2P_3 parallel to BC, P_3P_4 parallel to AB, P_4P_5 parallel to CA, and P_5P_6 parallel to BC. Here P_2, P_5 lie on AB; P_3, P_6 lie on CA; and P_4 on BC. Show that P_6P_1 is parallel to AB.

Exercise 2 (RMO 2004 P5,). Let $ABCD$ be a quadrilateral; X and Y be the midpoints of AC and BD respectively; and the lines through X and Y respectively parallel to BD, AC meet in O. Let P, Q, R, S be the midpoints of AB, BC, CD, DA respectively. Prove that

(a) quadrilaterals $APOS$ and $APXS$ have the same area;

(b) the areas of the quadrilaterals $APOS, BQOP, CROQ, DSOR$ are all equal.

Exercise 3 (RMO 2012a P5,). Let ABC be a triangle. Let BE and CF be internal angle bisectors of $\angle B$ and $\angle C$ respectively with E on AC and F on AB. Suppose X is a point on the segment CF such that $AX \perp CF$; and Y is a point on the segment BE such that $AY \perp BE$. Prove that $XY = (b+c-a)/2$ where $BC = a$, $CA = b$ and $AB = c$.

Exercise 4 (RMO 2014b P5,). Let ABC be a triangle with $AB > AC$. Let P be a point on the line AB beyond A such that $AP + PC = AB$. Let M be the mid-point of BC and let Q be the point on the side AB such that $CQ \perp AM$. Prove that $BQ = 2AP$.

Exercise 5 (RMO 2015c P8,). The length of each side of a convex quadrilateral $ABCD$ is a positive integer. If the sum of the lengths of any three sides is divisible by the length of the remaining side then prove that some two sides of the quadrilateral have the same length.

Exercise 6 (RMO 1992 P6,). $ABCD$ is a quadrilateral and P, Q are mid-points of CD, AB respectively. Let AP, DQ meet at X, and BP, CQ meet at Y. Prove that

$$\text{area of } ADX + \text{area of } BCY = \text{area of quadrilateral } PXQY.$$

Exercise 7 (RMO 1997 P1,). Let P be an interior point of a triangle ABC and let BP and CP meet AC and AB in E and F respectively. If $[BPF] = 4, [BPC] = 8$ and $[CPE] = 13$, find $[AFPE]$. (Here $[\cdot]$ denotes the area of a triangle or a quadrilateral, as the case may be.)

Exercise 8 (RMO 2003 P5,). Suppose P is an interior point of a triangle ABC such that the ratios

$$\frac{d(A, BC)}{d(P, BC)}, \frac{d(B, CA)}{d(P, CA)}, \frac{d(C, AB)}{d(P, AB)}$$

are all equal. Find the common value of these ratios. [Here $d(X, YZ)$ denotes the perpendicular distance from a point X to the line YZ.]

Exercise 9 (RMO 2010 P1,). Let $ABCDEF$ be a convex hexagon in which diagonals AD, BE, CF are concurrent at O. Suppose $[OAF]$ is geometric mean of $[OAB]$ and $[OEF]$ and $[OBC]$ is geometric mean of $[OAB]$ and $[OCD]$. Prove that $[OED]$ is the geometric mean of $[OCD]$ and $[OEF]$. (Here $[XYZ]$ denotes are of $\triangle XYZ$)

Exercise 10 (RMO 2012a P1,). Let ABC be a triangle and D be a point on the segment BC such that $DC = 2BD$. Let E be the mid-point of AC. Let AD and BE intersect in P. Determine the ratios BP/PE and AP/PD.

Exercise 11 (RMO 2012b P5,). Let ABC be a triangle. Let D, E be points on the segment BC such that $BD = DE = EC$. Let F be the mid-point of AC. Let BF intersect AD in P and AE in Q respectively. Determine BP/PQ.

Exercise 12 (RMO 2012c P5,). Let ABC be a triangle. Let D, E be points on the segment BC such that $BD = DE = EC$. Let F be the mid-point of AC. Let BF intersect AD in P and AE in Q respectively. Determine the ratio of the area of the triangle APQ to that of the quadrilateral $PDEQ$.

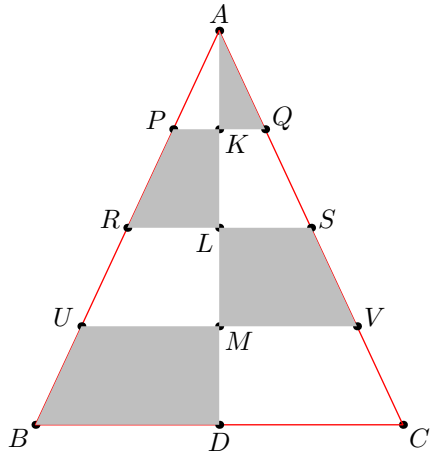


Figure 1: India RMO 2014a P1, Exercise 15

Exercise 13 (RMO 2012d P5,). Let ABC be a triangle. Let E be a point on the segment BC such that $BE = 2EC$. Let F be the mid-point of AC. Let BF intersect AE in Q. Determine BQ/QF.

Exercise 14 (RMO 2013e P4,). In a triangle ABC, points D and E are on segments BC and AC such that $BD = 3DC$ and $AE = 4EC$. Point P is on line ED such that D is the midpoint of segment EP. Lines AP and BC intersect at point S. Find the ratio BS/SD.

Exercise 15 (RMO 2014a P1,). In acute $\triangle ABC$, let D be the foot of perpendicular from A on BC. Consider points K, L, M on segment AD such that $AK = KL = LM = MD$. Suppose the sum of the areas of the shaded region equals the sum of the areas of the unshaded regions in Fig. 1. Prove that $BD = DC$.

Exercise 16 (RMO 2015c P1,). Let $ABCD$ be a convex quadrilateral with $AB = a$, $BC = b$, $CD = c$, and $DA = d$. Suppose $a^2 + b^2 + c^2 + d^2 = ab + bc + cd + da$ and the area of $ABCD$ is 60 square units. If the length of one of the diagonals is 30 units, determine the length of the other diagonal.

Exercise 17 (RMO 2015a P1,). Two circles Γ and Σ, with centers O and O', respectively, are such that O' lies on Γ. Let A be a point on Σ, and let M be the midpoint of AO'. Let B be another point on Σ, such that $AB \parallel OM$. Then prove that the midpoint of AB lies on Γ.

References

[**Che25**] Evan Chen. *The OTIS Excerpts*. Available at https://web.evanchen.cc/excerpts.html. 2025, pp. vi+289 (cited p. 1)